

Polydisperse structure factors for small-angle scattering: numerical Ornstein–Zernike solutions with arbitrary closures, and an assessment of the common decoupling approximations

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Abstract

Small-angle scattering from concentrated dispersions requires a structure factor, and real dispersions are polydisperse. Because the exact treatment — solving the coupled multi-component Ornstein–Zernike (OZ) equations for all $p(p+1)/2$ partial structure factors — has been considered expensive, most analysis packages instead use one of several decoupling approximations whose error is rarely quantified for the system at hand. We describe an implementation in which the exact solution is cheap enough to be the default: size polydispersity is discretised by moment-matched quadrature, so that three to five classes reproduce a continuous distribution, and the resulting matrix OZ problem is solved with the same accelerated fixed-point machinery used for the one-component case. Arbitrary pair potentials are made polydisperse without re-deriving them, by evaluating any one-component potential at the reduced separation r/σ_{ij} . We show that the structure factor and the form-factor average require different discretisations — a point that, if overlooked, produces spurious form-factor oscillations easily mistaken for physical structure — and that six commonly used approximation schemes can be compared directly against the exact result within a single interface. The implementation is validated against published Monte Carlo comparisons, against analytic Percus–Yevick hard spheres, and against an independent integral-equation library, and is distributed both as a Python package with a graphical interface and as a C plugin for SASfit.

Keywords: small-angle scattering; structure factor; polydispersity; Ornstein–Zernike; integral equation theory; closure relations

Synopsis: A general solver for the multicomponent Ornstein–Zernike equations is described, giving exact partial structure factors for polydisperse interacting particles under any of twenty-four closure relations and any of nineteen pair potentials, together with a quantitative assessment of the approximations normally used to combine a structure factor with a size distribution.

1 Introduction

The measured intensity of a concentrated dispersion of interacting particles does not factorise into a form factor and a structure factor unless the particles are identical. For a polydisperse system the exact expression involves the partial structure factors $S_{ij}(Q)$ of every pair of size classes,

$$I(Q) = \sum_{ij} \sqrt{\rho_i \rho_j} F_i(Q) F_j(Q) S_{ij}(Q), \quad (1)$$

and obtaining those requires solving the multicomponent OZ equations. In practice this is usually avoided in favour of one of several approximations — the monodisperse approximation, the decoupling approximation of Kotlarchyk & Chen [1], the local monodisperse approximation of Pedersen [2], a partial-structure-factor scheme, a scaling approximation, or a van der Waals one-fluid treatment. Each is well founded in some regime, but a user fitting data has no easy way to know which regime they are in.

Two developments make the exact route practical. First, the reduction of a continuous size distribution to a small number of discrete classes by *moment matching* rather than histogramming: D’Aguanno and Klein [3, 4] showed that three classes chosen as the nodes of a Gaussian quadrature reproduce results indistinguishable from five up to 30 % polydispersity, because the thermodynamics depends only on the low-order moments of the distribution. Second, the availability of robust accelerated fixed-point solvers, which make each OZ solve a fraction of a second even for the $p(p+1)/2$ coupled pair functions.

We describe an implementation that takes the exact solution as the default and exposes the approximations for comparison rather than as substitutes. The software extends the OZ solver toolkit distributed with SASfit [5, 6].

It is worth being precise about what is and is not new here, since the separate ingredients are all established. Analytic multicomponent structure factors have been available for decades — Vrij’s hard-sphere mixture solution [7], Blum and Høye’s multi-Yukawa mixture [8], and the polydisperse charged and adhesive models of Gazzillo and co-workers [9, 10] — and are implemented in several packages, notably `mixscatter` for the Vrij solution. General-purpose integral-equation solvers likewise exist, of which `OrnsteinZernike.jl` [11] is the most complete recent example, covering a wide range of closures for one-component systems. Small-angle scattering packages, for their part, supply the approximations: `SasView` and `jscatter` both implement decoupling and local monodisperse schemes alongside analytic structure factors.

What has been missing is the combination. The analytic solutions are exact but confined to particular potentials; the general solvers handle arbitrary closures but are not built around polydispersity or around scattering observables; and the scattering packages offer the approximations without a way to check them. A user fitting a concentrated polydisperse dispersion therefore chooses an approximation on the basis of custom rather than evidence. The contribution here is to make the exact multicomponent calculation the default for an arbitrary potential and closure, to compute the approximations alongside it so that the error of each is visible at the state point actually being fitted, and to validate the result against every analytic solution available rather than against itself.

Draft note: the paragraphs above place this work against the packages actually used during its development and validation, which is not the same as a systematic survey of the field. Before submission, check in particular whether any package now combines exact multicomponent solution with arbitrary closures, and expand the discussion of prior polydisperse OZ work beyond D’Aguanno & Klein. The natural starting reference remains Kohlbrecher and Breßler [6], which surveys the analytic-solution landscape.

2 Theory

2.1 Multicomponent Ornstein–Zernike equations

For p species with number densities ρ_i the OZ relation [12] couples all pair functions. In Fourier space, with $\mathbf{C}(Q)$ the matrix of $\hat{c}_{ij}$ and $\boldsymbol{\rho} = \text{diag}(\rho_i)$,

$$\mathbf{H}(Q) = [\mathbf{I} - \mathbf{C}(Q)\boldsymbol{\rho}]^{-1} \mathbf{C}(Q), \quad \boldsymbol{\Gamma} = \mathbf{H} - \mathbf{C}. \quad (2)$$

The single scalar division of the one-component problem is replaced by a small dense linear solve at each of the N radial grid points. Every closure relation that is an elementwise function of

Γ — Percus–Yevick [13], hypernetted chain, Rogers–Young [14], Verlet and the empirical bridge-function family — then applies unchanged to the matrix-valued $\Gamma_{ij}(r)$. Standard accounts of the closures and of integral-equation theory generally are given by Hansen and McDonald [15].

Three structure-factor conventions occur in this field and must be distinguished: the number-fraction form $S_{ij} = x_i \delta_{ij} + n x_i x_j \hat{h}_{ij}$ used by D’Aguanno and Klein [3], the Ashcroft–Langreth form $S_{ij} = \delta_{ij} + \sqrt{\rho_i \rho_j} \hat{h}_{ij}$ required by equation (1), and the measured single curve $S^M(Q)$. The conversion between the first two is a factor $\sqrt{x_i x_j}$; confusing them produces an error in $I(Q)$ proportional to the number of size classes, which is easily mistaken for a physical effect (§5.4).

2.2 Discretising the size distribution

A p -point Gaussian rule is exact for the first $2p - 1$ moments of the weight function. Since $I(Q \rightarrow 0)$ is governed by $\langle R^3 \rangle^2$ and $\langle R^6 \rangle$, exactness in the low-order moments is precisely what is required, and this is why three classes suffice where an equally spaced histogram needs seven to nine (Figure 1).

All five distributions implemented (Schulz, Gaussian, log-normal, Weibull, gamma) have analytic moments, but the map from moments to quadrature nodes — the Golub–Welsch Hankel eigenproblem — is classically ill-conditioned. Table 1 gives measured condition numbers for the log-normal case built from *exact* moments; they exceed double precision by $N \approx 10$ for a broad distribution, and, importantly, the Cholesky factorisation does not fail there but returns meaningless nodes. We therefore use a closed-form classical rule wherever one exists — generalised Gauss–Laguerre for the Schulz/gamma weight and Gauss–Hermite for the Gaussian, both stable in double precision — and reserve high-precision Golub–Welsch for the log-normal and Weibull cases, which have no classical rule. All four are moment-exact to $\sim 10^{-16}$.

Table 1: Condition number of the log-normal Hankel matrix built from exact moments. Double precision carries about 10^{16} ; past that the Cholesky factorisation does not fail, it silently returns nonsense.

	$s = 0.3$	$s = 0.4$	$s = 0.5$
$N = 6$	1.6×10^7	1.7×10^7	1.2×10^8
$N = 8$	2.1×10^{10}	1.9×10^{11}	4.7×10^{13}
$N = 10$	7.0×10^{13}	1.7×10^{16}	5.4×10^{19}
$N = 12$	6.4×10^{17}	1.2×10^{21}	2.3×10^{27}

The Gaussian rule requires a physical safeguard. Gauss–Hermite places its outermost node at $u \approx \pm 2.86$, so σ reaches zero near $s \approx 0.35$. For a form-factor average a vanishing bin is harmless; for a structure factor it is not, because a hard core must be placed at every σ_{ij} . Nodes below $10^{-3} \langle \sigma \rangle$ are therefore discarded and the weights renormalised, sacrificing moment-exactness rather than physicality. A test on $\sigma > 0$ alone is insufficient: at $s = 0.35$ the offending node lies at 10^{-4} .

2.3 Making an arbitrary potential polydisperse

Rather than re-deriving each pair potential for a mixture, we adopt

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j), \quad u_{ij}(r) = u(r/\sigma_{ij}), \quad (3)$$

so that every pair experiences the same interaction *shape* measured in units of its own contact distance. This is the assumption Robertus *et al.* [16] make for size-independent stickiness, and it makes the construction potential-agnostic: well depths, widths and exponents retain their reduced-unit meaning for every pair. Implementationally, each existing one-component potential

routine is evaluated on a rescaled radial grid, so no potential is re-implemented. With one size class the construction reproduces the original one-component code *bit-identically*.

A caution that cost us dearly, and which we report because the failure mode is general: a potential routine that positions its hard core by *array index* rather than against the radial grid never sees the rescaling, and then every pair receives its core at the same radius — the mixture is silently solved as though all particles were identical. See §4.

Equation (3) is a modelling choice, not a derivation, and it is not universally appropriate. For Lennard-Jones the conventional alternative is Lorentz–Berthelot mixing with per-species ε . More importantly it fails for *charge-coupled* potentials, where the amplitude scales with particle size ($Z_i \propto \sigma_i^n$) and the screening length is a global property of the whole distribution through the counterion density,

$$\kappa^2 = 4\pi L_B \sum_i n_i Z_i. \quad (4)$$

Note the first power of Z : screening is by the small ions. Charged systems are therefore handled by a dedicated construction following D’Aguanno and Klein [4], with the size dependence of the valence exposed as a user choice, since it is genuinely unsettled — constant surface charge density gives $Z \propto \sigma^2$, whereas charge-renormalisation saturation and electrophoresis on highly charged low-salt spheres both indicate $Z \propto \sigma$. The choice matters: it changes $g(r)$ by an amount comparable to the effect of polydispersity itself.

2.4 Structure factor and form factor need different discretisations

$S_{ij}(Q)$ varies smoothly with particle size, so a few moment-matched classes suffice, and since the OZ solve costs $O(p^2)$ transforms it is desirable to keep p small. The form-factor average $\langle |F|^2 \rangle$ behaves quite differently: at high Q the phase QR spans approximately $Q\langle\sigma\rangle s$ radians across the distribution, so a handful of classes superposes a handful of ringing form factors instead of smearing them. No amount of moment-exactness repairs this, because the integrand is not polynomial in R .

We therefore decouple the two, solving the OZ equations on p_S classes and evaluating the intensity sum on a denser grid p_F , with the rule of thumb $p_F \gtrsim Q_{\max}\langle\sigma\rangle s$. Only the smooth part of the structure factor may be interpolated onto the denser grid: writing $S^{\text{AL}} = \delta_{ij} + \sqrt{\rho_i \rho_j} \hat{h}_{ij}$, interpolating the whole matrix smears the Kronecker delta into a band and converts the incoherent sum $\sum_i |F_i|^2$ into the coherent $|\sum_i F_i|^2$ — which is precisely what oscillates.

A separate and equally silent requirement is that the radial grid must resolve the *narrowest feature of the potential*, not the particle: for a well of width δ one needs of order $30\sigma/\delta$ points per diameter (§4.6).

2.5 Instrumental resolution

Small-angle neutron scattering resolution $\Delta Q/Q$ is of order 10% and smears form-factor oscillations in exactly the way polydispersity does. Fitting without it lets the model absorb the instrument into the fitted width. On smeared synthetic data with a true relative width of 0.15, a fit that ignores resolution returns 0.1726 — 15% too large — with $\chi_{\text{red}}^2 = 1.07$; including it returns 0.1497, within 0.2%. The point that matters is that the unsmeared fit does not *look* bad: this is bias, not imprecision, and more data does not remove it.

Smearing is applied as a fixed linear operator built once from $(Q, \Delta Q)$. For radially averaged data the correct kernel is the azimuthally integrated one of Pedersen et al. [17], which is *not* Gaussian; their Gaussian expression is its large-argument limit. The difference is confined to the lowest few points of a run, where collimation dominates and $\sigma/Q \rightarrow 1$: there the centre of mass of the true kernel sits 59% above $\langle Q \rangle$ against 15% for the Gaussian. It costs nothing to use the correct form, since completing the square turns the product of a vanishing exponential and a diverging Bessel function into a numerically benign expression.

Smearing is inexpensive here for a structural reason: the extended, denser grid it requires would normally multiply the model cost several-fold, but the expensive part — the OZ solve — does not depend on the Q grid at all. It also leaves the analytic elimination of scale and background intact, since smearing is linear and the flat instrumental background is not smeared.

3 Implementation

The package is written in Python (NumPy/SciPy) with a Tk interface, and is also available as a C plugin for SASfit. Nineteen closure relations are available for polydisperse systems; four (reference HNC, modified HNC, rescaled MSA and the Euler–Rahman closure) require a one-component reference solve or a hard-sphere-specific construction and are excluded rather than offered and allowed to fail.

Fixed-point acceleration follows the same strategy in both implementations: Anderson acceleration [18] (SUNDIALS KINSOL KIN_FP) is tried first, with Newton–Krylov (GMRES with line search) and Picard iteration as successive fallbacks. Anderson converges from a cold start ($\Gamma = 0$) at every state point we have tested, including hard spheres at $s = \varphi = 0.40$, and is three to six times faster than Picard, the gain growing with the difficulty of the point.

The Picard fallback is implemented in the averaged (Mann, or Krasnoselskii–Mann) form [19]

$$\Gamma_{n+1} = (1 - \alpha) \Gamma_n + \alpha T(\Gamma_n), \quad (5)$$

with $\alpha = 1$ — undamped Picard — as the default. The averaging is not cosmetic: for a nonexpansive map the Krasnoselskii–Mann theorem gives convergence where $\alpha = 1$ need not converge at all, and we find state points where the distinction is decisive. A Lennard-Jones fluid at $\varepsilon = 0.8 k_B T$, $\varphi = 0.30$ under HNC diverges at $\alpha = 1$ and converges cleanly at $\alpha = 0.5$ on one discretisation, while on another the two exchange roles. The fixed point there is simply near the edge of stability, and no single choice of α is right everywhere.

3.1 Verification by independent solvers

All available solvers reach the *same* fixed point on well-behaved states, agreeing in $I(Q)$ to 10^{-11} or better, so the choice is normally one of cost rather than of answer. That is not something to be assumed, however, and the reason is structural rather than a matter of implementation quality.

The Ornstein–Zernike equation with a given closure is a nonlinear integral equation whose *number* of solutions changes with density and temperature. Beardmore et al. [20] analysed its bifurcation structure for the Lennard-Jones potential under HNC and showed that a subcritical isotherm carries at least one *fold bifurcation* in density, and that pairs of solution branches collide at what they term pseudospinodals — one branch having positive compressibility and the other negative, the latter being physically meaningless. They note explicitly that this explains “previous inconclusive attempts to compute solutions using Newton–Picard methods that are popular in the physical chemistry literature”. Peplow et al. [21] show further that solutions exist inside the so-called no-solution region which direct Newton or Picard iteration cannot reach at all, and recover them by pseudo-arclength continuation.

We observe precisely this. At a Lennard-Jones state well inside that region ($k_B T/\varepsilon = 1.25$, $\rho\sigma^3 = 0.57$), nine available solvers divide cleanly into two groups according to their *algorithm family*. Five fixed-point methods — SUNDIALS KINSOL’s fixed-point iteration, damped Picard, two Anderson implementations and a Biggs–Andrews scheme — all return $g_{\max} = 2.1636$ with $\min S(Q) = +0.21$. Four Newton–Krylov methods, three of them SUNDIALS variants differing only in the linear solver, instead return $g_{\max} = 1.13$ and 1.22 , every one with $\min S(Q)$ near -38 . All nine have residuals below 10^{-11} , so these are not failed calculations: they are converged solutions on the negative-compressibility branches the bifurcation analysis identifies.

That the split follows the algorithm family rather than the implementation is what one would expect from the analysis. At a fold the Jacobian is singular, so a Newton-based method is drawn away from the physical branch precisely where the fold lies, while a fixed-point iteration is not. The distinction is not as sharp as it looks — Walker and Ni [22] show that Anderson acceleration applied to a stationary iteration is essentially GMRES on the equivalent left-preconditioned system, so an accelerated fixed-point method is closer to a Krylov method than its name suggests — but the empirical separation here is complete. A solver reporting success is therefore not necessarily wrong about having found *a* root; it may simply have found a different one, and which one depends on how it looks.

Nor are these the only solutions. Applying the deflation technique of Farrell et al. [23] — which eliminates a known root from the residual so that a solver started from the same point must converge elsewhere — yields a further solution at $g_{\max} = 1.476$, again with $\min S(Q) < 0$ and a residual of 4×10^{-10} . At least four distinct fixed points therefore exist at this state, of which one is physical. We make no claim to have enumerated them: deflation finds distinct solutions where a solver converges, and does not certify that none remain.

Two consequences follow for software intended to be used unattended. First, the physicality test is not a diagnostic nicety but the discriminator between branches: since a structure factor is a variance it cannot be negative, and the condition $\min S(Q) \geq 0$ — the same criterion used to reject branches in the bifurcation studies — removes every one of the unphysical solutions above. We therefore apply it unconditionally rather than as a warning. Second, since a small residual is necessary but not sufficient, the library verifies by default: every solve is repeated with an independent solver from a *cold start* and the result returned only if the two agree.

The choice of verifier deserves more care than it first appears to. The obvious instinct is to pick a solver from a different algorithm family, on the grounds that it cannot share a failure mode. The measurements above say otherwise: a Newton–Krylov verifier would report disagreement at every state near a fold, but because the *verifier* had left the physical branch rather than the solver under test. We therefore verify with a second fixed-point method — a different implementation rather than a different algorithm class — and recommend the same to anyone building a similar check.

The cold start is essential and cannot be optimised away. Warm-starting the verifier from the first solver’s answer is roughly three times faster and destroys the check: a wrong root is still a root, so a solver placed on it sees a vanishing residual and stays there. Started from one of the erroneous roots above, a solver that finds the physical branch unaided stays on the erroneous one instead, the two “agree”, and the wrong answer is certified. What the check must ask is whether independent searches from different starting points reach the same basin, and that requires the starting points to be independent.

We emphasise that none of this makes the solvers defective. The branches are real, their existence is established, and no fixed-point or Newton method can be expected to choose between them unaided. Where a branch must be followed through a fold — rather than merely detected — pseudo-arclength continuation is the appropriate tool, and we do not currently implement it. That is a limitation of this software, not of the analysis.

A further caution applies to the solvers’ own reporting, and it cuts both ways. KINSOL was observed to return success after diverging, because the exponent clipping in the closure pins values at $\sim 10^{217}$ and defeats its internal stopping test; and a damped Picard iteration whose convergence test compared successive norms of Γ , rather than the change in Γ , reported a converged solution at a point whose residual was $\|T(\Gamma) - \Gamma\| = 19.4$. Every solve therefore recomputes the residual before accepting the result, and we would urge the same of any comparable code: a convergence flag is a claim, whereas the residual is evidence.

That distinction is what separates the two situations we have described. For the Lennard-Jones state above, every one of the nine solvers returns a point whose residual is below 10^{-11} : the five fixed-point methods converge to the physical branch and the four Newton–Krylov methods

converge, equally legitimately, to branches with $\min S(Q)$ near -38 . Those are genuinely distinct solutions. By contrast, at a strongly coupled state of the Ballone–Pastore–Galli–Gazzillo closure, eight of nine solvers agree to 10^{-12} on a single root and the ninth — the undamped Picard iteration with the defective stopping test — returns a different value that is not a solution at all. Without the residual the two cases are indistinguishable, and we would have reported a bifurcation where there was only an unconverged iteration.

Finally, a solver’s cost can be dominated by a single unset option, which is another reason to re-measure rather than assume: `scipy`’s Anderson implementation was by a wide margin the slowest available — 30 s where the others took 0.5 s — purely because the initial inverse-Jacobian scale was left at its default. Correcting it makes that solver competitive, a factor of 15 to 66 depending on the state point, with the fixed point unchanged to 10^{-9} . A timing and agreement comparison across solvers is worth repeating whenever a new potential or closure is added, and the software provides one.

The multiple-fixed-point problem is not confined to the Lennard-Jones example: at one strongly coupled state two different accelerated solvers converged to residuals of $\sim 10^{-12}$ and returned $S(Q) = 17.0$ and 9.7 respectively, both with $\min S(Q) < 0$.

3.2 Cost

The claim that the exact treatment is cheap enough to be the default should be supported by numbers rather than asserted. Those below were measured on an Intel Core i7 (Tiger Lake) under Windows 11 with Python 3.14 and NumPy 2.5, using the benchmark script distributed with the package. Each figure is the best of several repeats.

Table 2: Time per Ornstein–Zernike solve, hard spheres at $\varphi = 0.3$. The exact multicomponent calculation on five size classes costs a few seconds; a solver that is merely available rather than chosen can cost an order of magnitude more.

Size classes p	time (s)	Solver	time (s)
1	0.06	SUNDIALS KINSOL, fixed point	0.0049
3	1.53	<code>scipy</code> Anderson	0.0084
5	3.04	Anderson acceleration	0.0100
7	5.77	SUNDIALS KINSOL, GMRES	0.0107
10	10.84	<code>scipy</code> Newton–Krylov	0.0224
		Biggs–Andrews	0.0405
		Picard iteration	0.0542

Three points are worth drawing out, two of which qualify claims made earlier in this paper.

The approximation schemes are not uniformly free. Once Γ has converged, the monodisperse, decoupling and local monodisperse approximations cost 0.0013–0.0026 s against 0.022 s for the exact intensity — around a tenth, so computing them alongside it is genuinely negligible. The partial-structure-factor, scaling and van der Waals schemes cost 0.078–0.088 s, three to four times the exact calculation, because each is a double sum over the p_F form-factor classes rather than a single pass. The argument for computing every scheme beside the exact result holds for half of them.

The scaling with the number of classes is milder than p^2 . The matrix problem involves $p(p+1)/2$ pair functions, which suggests quadratic growth, but from $p = 3$ to $p = 10$ the measured time rises by a factor of 7.1 for a threefold increase in p , an exponent near 1.6. The discontinuity between $p = 1$ and $p = 3$ — a factor of twenty-five — is larger still, and reflects the one-component path avoiding the matrix solve altogether.

Verification is the dominant cost, by construction. A verified solve takes 0.044 s against 0.0048 s unverified, a factor of nine. That ratio is unflattering precisely because the primary solver is fast: the verifier is deliberately an independent fixed-point method rather than the quickest available one (§3.1). We regard this as the right trade for software used unattended, but it is a real cost, and verification can be disabled where a state point is known to be well behaved.

For the quantity a user actually waits on, a three-parameter fit to eighty data points — radius, polydispersity and volume fraction, roughly thirty solves — completes in 19 s to $\chi_{\text{red}}^2 = 0.75$. Grid resolution enters mildly over the usual range, 0.0026 s at fifty points per diameter rising to 0.047 s at eight hundred, so the narrow-well systems of §4.6 remain affordable despite needing an order of magnitude more points than a hard sphere.

3.3 Programmatic and graphical interfaces

A complete calculation is a single call:

```
result = ozLib.solve("HardSphere", phi=0.3,
    closure="Rogers-Young", findConsistentParameter=True,
    solver="SUNDIALS KIN_FP")
```

Closures and solvers are data, not code paths: both are dictionaries mapping a label to a setter, so adding a closure requires a bridge-function expression, a setter and one dictionary entry, after which the graphical interface, the consistency search and the polydisperse machinery all pick it up without modification. Potential and closure parameter fields in the interface are generated by introspection of the corresponding setter (Figure 4), so a new potential appears with correctly named and correctly defaulted fields without any interface code being written.

Setting `findConsistentParameter` determines the free parameter of Rogers–Young, HMSA [24], BPGG and the other one-parameter closures by requiring the compressibility and virial routes to the pressure to agree, rather than requiring a value from the user. For a mixture this is a single scalar condition,

$$\chi^{-1} = 1 - n \sum_{ij} x_i x_j \hat{c}_{ij}(0) = \frac{\partial(\beta P)}{\partial n}, \quad (6)$$

which avoids the full compressibility-matrix problem. The virial pressure retains the hard-core contact term; D’Aguanno and Klein [4] omit it legitimately, their macroions being too strongly charged to reach contact, but it is first order in the pressure for a general potential with a reachable core. A consistent parameter does not always exist — the residual is frequently monotone in the mixing parameter and already non-zero in the Percus–Yevick limit — and this is reported rather than concealed behind a nearest-endpoint fallback presented as a fit.

4 Validation

The implementation was checked against analytic solutions, against published tabulations, and against four independently written software packages. We report the comparisons in full, including the two cases in which they uncovered defects in our own code, because the pattern is itself informative: every defect found late in this work was invisible to internal consistency checks and was exposed only by an external reference.

4.1 Analytic and literature references

4.2 Comparison with independent software

An absolute check: the compressibility. Every comparison above sets one calculation against another. One quantity admits an exact answer instead. For monodisperse hard spheres

Table 3: Comparison with analytic results and published tabulations. Where a difference is quoted it is the deviation from the reference, not from a previous version of this code.

Reference	Quantity	Ours	Agreement
Wertheim [25], Thiele [26]	$c(r)$, PY hard spheres	numerical	first order in Δr
	$g(\sigma^+)$, $\eta = 0.1\text{--}0.4$	numerical	0.1 %
Analytic PY, $\eta = 0.3$	$S(0) = (1 - \eta)^4 / (1 + 2\eta)^2$	numerical	converges
	χ^{-1}	11.07	10.66 (extrapolation)
	$\beta P / \rho$	3.890	3.816 (grid)
	PY inconsistency	+1.33	+1.25
D’Aguanno & Klein (1991)	$1/S(0)$, excess pressure	—	0.01 %, 0.02 %
	excess energy, $g_{\max}$	—	0.27 %, 0.54 %
Vrij [7] mixture PY	S_{ij}^{AL} , 3 classes	numerical	0.09 %

Table 4: Comparison with independently written packages. “exact” denotes agreement to all printed digits; the remaining entries are limited by discretisation and converge as Δr is reduced.

Package	Model	Method	Agreement
<code>mixscatter</code> [27]	mixture PY [7]	analytic	0.09 %
<code>jscatter</code>	one-component PY	analytic	0.02–0.12 %
<code>jscatter</code>	RMSA [28]	analytic	see §4.2
<code>two_yukawa_reference</code>	two-Yukawa MSA [29]	analytic	0.21 %
<code>jscatter</code>	adhesive hard spheres	analytic	see §4.6
<code>jscatter</code>	OZ solver, PY and HNC	numerical	0.2–0.4 %
<code>OrnsteinZernike.jl</code>	4 bridge functions	analytic	$\sim 10^{-16}$

the Percus–Yevick compressibility is

$$S(0) = \frac{(1 - \varphi)^4}{(1 + 2\varphi)^2}, \quad (7)$$

so extrapolating the computed partials to the origin and comparing measures absolute accuracy rather than agreement.

The result is that the transform type matters far more than its place in a cost table suggests. At $\varphi = 0.40$ and 100 radial points per diameter, the first-order DST-I gives $S(0)$ low by 5.27 %; the second-order DST-IV is low by 0.036 % on the same grid, and by 0.004 % at four times the resolution. Type 1 refines as 3.96 per fourfold step, textbook first order, so it reaches 1.33 % where type 4 began 150 times better.

The cause is where the discontinuity lands. A DST-I places nodes at $r = (n + 1)\Delta r$, so the hard core at $r = \sigma$ falls ON a node and is carried by a point that is neither inside nor outside; the excluded volume is then wrong at first order in Δr . A DST-IV places them at $(n + \frac{1}{2})\Delta r$, the core falls between points, and the step is resolved to second order.

Two things make this worth reporting rather than absorbing into a recommendation. First, the error is independent of the real-space range: extending $r_{\max}$ from 41σ to 655σ left $S(0)$ unchanged in the sixth decimal. A convergence study that refines resolution sees orderly first-order behaviour and concludes the method works — which it does, slowly, towards an answer whose leading error is set by grid alignment. Second, $S(0)$ is the isothermal compressibility, so this is a systematic bias in exactly the quantity the volume fraction controls, and a fit will move φ to absorb it.

It also accounts for a discrepancy that had resisted explanation: a multicomponent solve converged to a fixed point at 7×10^{-12} yet differing from two independent analytic references by $\sim 10^{-3}$, the difference immune to refinement and growing with φ . That is the same effect seen from the other side.

The practical consequence is a restriction rather than a fix. Type 4 needs every pair core between grid points, which one size class can be given and a mixture cannot: the pair diameters $\sigma_{ij} = (\sigma_i + \sigma_j)/2$ come from a moment-matched quadrature and are mutually incommensurate. The polydisperse route is therefore first order, and until a mixture-capable second-order treatment exists a dense polydisperse fit should use the finest affordable resolution, with its fitted volume fraction read accordingly.

Reproducibility of the figures and of the table. Three gaps are recorded here rather than left to be discovered.

The remaining `jscatter` rows above — one-component PY, RMSA, adhesive hard spheres and the numerical OZ comparison — were measured where `jscatter` was installed. Its dependency chain does not build under MinGW, so they cannot be re-run on the machine this work is written on. The two-Yukawa row was brought back within reach by extracting that solver into `tools/two_yukawa_reference.py`, which needs `numpy` alone; the others deserve the same treatment before submission.

`fig3_oscillation.pdf` has no generator in the repository. The figure exists and nothing can reproduce it, which means a referee asking for any change to it would find there is nothing to change. `fig_survey.pdf` names `tools/section53_survey.py` in its own caption and is the model the others should follow; `fig1_discretisation.pdf` now has `tools/fig1_discretisation.py`.

Regenerating it also caught a discrepancy in the figure itself. `polydisperse_nodes` offers two discretisations — `sizeClasses`, the moment-matched generalised Gauss–Laguerre rule the caption describes, and `quantileClasses`, which is quantile-based — and only the first reproduces the moments exactly. At $p = 3$ the Gauss–Laguerre rule returns $\langle \sigma \rangle = 1.000000$ and $\langle \sigma^2 \rangle = 1.062500$, exactly $1 + s^2$; the quantile rule returns 0.9794 and 0.9596. Both are usable discretisations, but only the first supports the claim printed beside the figure. The moments are now shown on each panel so that the claim can be checked rather than taken on trust.

The table’s figures depend on the transform, and the table does not say so. The one-component Percus–Yevick row is quoted as 0.02–0.12 % over $0.2 \leq \varphi \leq 0.4$, and that is a DST-IV result. With the first-order DST-I at the same 100 radial points per diameter the same comparison gives 0.70, 1.37 and 2.33 % at $\varphi = 0.2, 0.3$ and 0.4 — an order of magnitude worse, rising with density, and reducible only fourfold per fourfold refinement (0.17, 0.33, 0.51 % at 400 points).

This is the grid-alignment effect described above — the hard core falling ON a DST-I node rather than between DST-IV nodes — seen in a second quantity, and it carries the same caveat: type 4 is available for ONE size class and not for a mixture, so the polydisperse route — which is what this paper is about — carries the first-order figures rather than the quoted ones. A reader comparing packages should know which transform produced a number before comparing it with their own.

RMSA: exact until the rescaling engages. This row was quoted as exact at all nine state points tested, and that was true of the points tested. It was not a useful claim, because at none of them did the Hayter–Penfold rescaling actually trigger: the rescale iteration count is zero throughout, so all nine agreements are for the unrescaled mean spherical approximation and the feature the method is named for went unexercised. Agreement to 10^{-16} on nine points that never invoke the mechanism under test is a weaker result than it appears, and the error was ours for choosing conditions gentle enough that plain MSA already returns a physical answer.

Repeating the comparison where the rescaling does engage — dilute, strongly charged and long-screened, which is the case Hayter and Penfold wrote it for, with the effective volume fraction lifted by factors of 2.3 to 34 — gives a different picture. Both implementations find the SAME rescaled volume fraction, agreeing to four decimals at eight of the nine points, so the rescaling

search itself is right. The resulting curves, however, agree only to between 10^{-14} and 3×10^{-5} , and the discrepancy grows systematically with coupling strength: 10^{-14} at $\gamma = 50$, 3×10^{-5} at $\gamma = 500$. That is the signature of an iterative tolerance inside the rescaling rather than of two different closed forms, and it is small enough to be irrelevant to any fit — but it is not exactness, and the distinction matters because both routes solve the same equations and should in principle agree to rounding.

One of the nine does not solve. At $\varphi = 0.1$, $\gamma = 50$ and a screening length of 10 particle radii the library returns “*NEWTON ITERATION NON-CONVERGENT while searching for the rescaled volume fraction*”, and the array returned alongside that status is the unrescaled result — meaningful to inspect, but not offered as an answer. The failure is reported clearly; it was our comparison script that ignored the status and compared the array regardless, producing a 168% “discrepancy” that was nearly written up as a silent fallback to MSA. That is the more instructive half of the episode: a check which discards the error code it is handed does not find defects, it manufactures them.

The message also explains the drift. The rescaling is not a closed form but a ROOT-FIND: Hayter and Penfold solve for the effective volume fraction at which the MSA solution becomes physical, and that condition is transcendental. Two implementations descending from the same algorithm converge to the same root — hence the four-decimal agreement in φ_{eff} — but stop at different points along the way, and the residual difference grows as the root becomes harder to locate, which is exactly the observed scaling with γ . Pushed further the iteration fails outright, and it does so in both codes: `jscatter` raises the same non-convergence at three conditions neighbouring the one that defeats ours.

That makes this the second method examined here whose domain is unmapped. The other is the Carbajal–Tinoco bridge function [30], which is implicit, $b = T(\gamma + b)$, and has no solution at all below a threshold: substituting $w = \gamma + b$ turns it into the explicit map $\gamma(w) = w - T(w)$, whose range is bounded below. Numerically $\gamma_{\text{min}} = -0.509$ at $\lambda = 0$, rising towards zero as λ grows, with two solutions above it and none below. Neither the source paper nor the comparison of Pihlajamaa and Janssen [11] locates that boundary; the latter simply excludes closures from its figures “due to convergence issues at this state point”.

In both cases the literature reports the method without stating where it stops working, and in both the boundary lies at conditions that are not exotic — here dilute and strongly charged, which is precisely the regime RMSA is used for in small-angle scattering, and there a contact value of γ routinely reached at moderate density. Locating such boundaries is open work, and the practical consequence for a user is that a closure or a rescaling failing to converge is not necessarily a sign of a bad state point or a bad implementation: it may be a region where the method has no answer to give.

4.3 Analytic references implemented in this work

Two of the three classical solvable polydisperse models had no reference implementation available to us, so we implemented them. Together with the Vrij hard-sphere solution supplied by `mixscatter`, they give exact analytic coverage of the polydisperse hard-sphere, charged-hard-sphere and sticky-hard-sphere fluids, against which the numerical solver can be checked at any state point.

Polydisperse charged hard spheres. Appendix A of Gazzillo, Giacometti & Carsughi (1999), the corrected form of their 1997 paper — the correction supplies a factor π omitted from P_z and repairs an error in the determinant. This is the same physics implemented by the FLAC program of Carsughi *et al.*, which cites the uncorrected 1997 paper. Our implementation reproduces the neutral monodisperse limit against analytic Percus–Yevick to 3×10^{-14} , the neutral polydisperse limit against `mixscatter` to 1×10^{-13} , and, for charged systems, all three effects the authors report for increasing polydispersity: the first peak is lowered ($1.7350 \rightarrow 1.3546$),

shifted to smaller Q ($5.841 \rightarrow 5.030$ in units of $Q(\sigma)$) and $S(0)$ increased ($0.0187 \rightarrow 0.0484$), reproducing their component counts (79, 112, 149) exactly.

One limitation of that route is worth stating, because it is a modelling assumption rather than an approximation that improves with effort. The corresponding-states construction takes the valence to be PROPORTIONAL TO SURFACE AREA, so charge polydispersity is fully correlated with size polydispersity: one distribution governs both. A sample whose surface chemistry does not scale with area — differing grafting densities across a size fraction, say — cannot be represented.

Ginoza and Yasutomi [31] treat size and interaction polydispersity as *independent*, on the same factorisable-coefficient mean-spherical solution, and would be the route to take if that case ever arose. Both descend from Ginoza [32], who showed that factorisable coefficients $K_{ij} = K d_i d_j$ collapse the multicomponent Blum–Høye equations to rational functions of a single scalar Γ — which is what makes any of this affordable for a size distribution, since each component then enters only through (σ_i, ρ_i, d_i) and the cost per evaluation is linear in the number of classes rather than quadratic.

A partial implementation of the Ginoza route exists in `ginoza_yukawa_msa.py` and is deliberately shelved: it solves for Γ and stops short of $S(q)$. Finishing it would reproduce, from equations transcribed out of a scanned 1986 paper, a structure factor the Gazzillo route already computes and which is validated above at machine precision. The two closing conditions Ginoza gives, his Eqs. (38a) and (38b), are both implemented there and agree to 10^{-12} , so the transcription is at least self-consistent; that is recorded in case the decoupled-charge case ever makes the module worth completing.

Polydisperse sticky hard spheres. Gazzillo and Giacometti [10], a mean-spherical solution in the Baxter sticky limit with factorisable coupling parameters $K_{ij} = Y_i Y_j$, which renders the Baxter factor matrix three-dyadic and collapses the $p \times p$ inversion to a 4×4 determinant for any number of components. Validation: the $\gamma_0 = 0$ limits against analytic Percus–Yevick (4×10^{-14}) and `mixscatter` (1×10^{-13}); the monodisperse sticky compressibility against their closed form (3×10^{-8}); the critical point $\eta_c = (\sqrt{3} - 1)/2$, $\gamma_{0c}^2 = (\sqrt{3} + 2)/6$ located as the divergence of $S(0)$; and the generalised Boyle temperatures of their Eq. (53) to the printed digits (2.787, 1.677, 2.986, 2.897 against 2.79, 1.68, 2.99, 2.90).

Note that the stickiness parameter of this model is *not* Baxter’s τ [33]: Baxter’s Percus–Yevick solution for mixtures requires $p(p+1)/2$ coupled quadratic equations and is not in general dyadic, which is why the mean-spherical model exists. The two agree only in the hard-sphere limit.

Three entries in Table 4 deserve comment.

The **RMSA** comparison was exact at all nine state points tested ($\eta = 0.05$ – 0.25 , screening lengths 2–10 nm, contact potentials 20–100 $k_B T$), including four at which the Hayter–Penfold rescaling [28] procedure engaged, so the agreement is not merely that of the underlying MSA branch. Two independently written implementations of a numerically delicate algorithm reproduce each other to all printed digits.

The **two-Yukawa** comparison validates the hard-core double-Yukawa potential, including cases of mixed sign (attractive K_1 , repulsive K_2), which is the regime in which a sign convention is most easily mistaken. The reference implementation states the identical convention, $V(r) = -k_B T [K_1 e^{-Z_1(r-1)} + K_2 e^{-Z_2(r-1)}]/r$ for $r > \sigma$.

The **numerical** comparison is the only external check available for the hypernetted-chain closure, which has no analytic solution. Both solvers are discretised and differ in transform, mixing scheme and hard-core representation, so the 0.3 % agreement is the expected order.

4.4 Two defects found by external comparison

Identical hard cores in mixtures. Comparison with `mixscatter` revealed a discrepancy of 5.2 % that, diagnostically, did *not* decrease under grid refinement and was *linear in the size*

spread, vanishing exactly in the monodisperse limit. The cause was that the hard-sphere potential routine positioned its core by array index rather than against the radial grid, so it did not see the rescaled grid on which the polydisperse construction of equation (3) relies. Every pair therefore received its core at the same radius: the mixture was being solved as though all particles were identical. After correction the agreement is 0.09 % and converges. It was the only one of twenty-one potentials written in that form.

4.5 Two numerical traps in fitting

Both of the following are properties of the fitting problem rather than of this implementation, and both produce a converged fit with a plausible χ^2 and silently wrong parameters. We report them because neither is evident from the literature and both cost us considerable time.

A rank-revealing linear solve can discard a parameter entirely. It is natural to split the fit: the scale, the flat background and any power-law amplitude enter the intensity linearly and can be obtained exactly by weighted least squares at every iteration, leaving only the shape parameters to the nonlinear optimiser. The saving is real. The hazard is that the design columns need not be comparable in magnitude. A model built from scattering length densities in cm^{-2} carries the contrast squared and so has magnitude $\sim 10^{20}$, while the constant background column is unity — a condition number near 10^{20} . `numpy.linalg.lstsq`, like any rank-revealing solver, discards singular values below $\max(M, N)\epsilon$, so the background and power-law coefficients are returned as *exactly* zero and the solve reports success. The symptom is a fit that a hand-chosen background beats by tens of percent in χ^2 , which is indistinguishable from a broken optimiser. Equilibrating the columns to unit norm before the solve and rescaling afterwards recovers all coefficients to five figures and changes nothing where the problem was already well conditioned [34].

The same scaling defeats the nonlinear optimiser: a parameter of order 10^{10} satisfies `least_squares`’s gradient test after a single evaluation, so it never moves and is reported with a meaningless uncertainty. Since the absolute scattering length densities are in any case perfectly correlated with the scale, we fit their `RATIO`, which is of order unity and whose sign is the quantity the data actually determine.

Gaussian resolution smearing fails where it is needed most. A small-angle instrument broadens Q in two dimensions in the detector plane, and the data are then radially averaged; the distribution of the magnitude about a true Q_0 is Rician, Eq. (8) below, not Gaussian [17, 35]. A Gaussian in Q is its $Q_0/\sigma \gg 1$ limit and is entirely adequate wherever dQ/Q is small.

$$P(Q|Q_0) = \frac{Q}{\sigma^2} \exp\left(-\frac{Q^2 + Q_0^2}{2\sigma^2}\right) I_0\left(\frac{QQ_0}{\sigma^2}\right) \quad (8)$$

It is not always small. For a spherical-shell dataset used to develop this work, dQ/Q reaches 0.55 at the lowest measured point, where a Gaussian kernel places 44 % of that point’s weight below $Q_{\min}$ and some of it at *negative* Q . The leading factor of Q in (8) suppresses the origin and excludes negative Q by construction, reducing that to 34 %. Both forms reproduce the unsmeared curve to 2×10^{-16} as $dQ \rightarrow 0$, which is the check that catches a mis-normalised kernel. Evaluate (8) through the exponentially scaled $\tilde{I}_0(x) = e^{-|x|}I_0(x)$: the argument reaches several hundred at these widths, where I_0 overflows a double long before the prefactor cancels it.

A related trap follows. A background term AQ^{-4+d} diverges at the origin while the smearing kernel reaches below the measured range, so the smeared value at the lowest point is governed by whatever guard prevents division by zero — in our case 6.6×10^5 times the unsmeared value, and varying by five orders of magnitude as that guard was moved. The power law describes the measured range and says nothing about $Q \rightarrow 0$, where any real curve turns over; holding it constant below the lowest measured Q removes the arbitrariness.

Draft note on citations. Now placed: OrnsteinZernike1914 and PercusYevick1958 (the equation and closure, §2), HansenMcDonald2013, Anderson1965, WalkerNi2011, HayterPenfold1981, Liu2005, DiazMaier2024 (mixscatter), BarkerPedersen1995 and GolubVanLoan2013.

Still in ozgui.bib and uncited: DuhHenderson1996, Gazzillo1999, Gazzillo2006, KalyuzhnyiCummings2006, Carsughi2000. Each has a natural home — the first in the discussion of mixing rules, the Gazzillo papers where stickiness and its size dependence are treated — but placing a citation where it is not actually used is worse than leaving it out, so they are left for whoever writes those sections.

Still absent and worth adding: software citations for NumPy, SciPy, SUNDIALS and NLOpt, all of which the work depends on.

One substantive point from the verification. Barker and Pedersen [35] conclude that a Gaussian approximation to the resolution is PREFERRED in most experiments for ease of calculation, and is inadequate only in particular cases — of which they name Porod scattering. The argument above should be framed as identifying one of those known exceptions rather than as a general correction, which is both more defensible and more useful.

4.6 What these figures measure

A note on what these figures measure. Every agreement quoted in this section was obtained with the type-1 discrete sine transform, whose grid points fall *on* the hard core, and which is therefore first order in Δr . A type-4 transform places the grid at $(n + \frac{1}{2})\Delta r$ so that the core lies between points — the midpoint rule — and is second order. For one component it is both more accurate and cheaper, and both advantages grow with resolution:

points per σ	type-4 time	accuracy gain
100	0.73×	139×
400	0.65×	547×
800	0.49×	1092×

The times are relative to type 1 at the same points per diameter; the accuracy gain is the ratio of the two errors. The growth of that ratio with resolution is the second-order against first-order signature. The figures quoted above would improve correspondingly.

The two transforms require different grid *lengths*, which is easy to overlook and costly to get wrong: a DST-I is computed through an FFT of length $2(N + 1)$ and so wants $N = 2^k - 1$, whereas a DST-IV is a length- N transform and wants $N = 2^k$. Sizing both for the former leaves the latter on a composite length ($16383 = 3 \times 43 \times 127$) and costs it a factor of three in the transform, enough to make the second-order method appear slower than the first-order one. Accuracy is unaffected by the choice.

We nevertheless retain type 1 as the default, because the type-4 condition cannot be met for a mixture. It requires every pair core σ_{ij} to fall between grid points, and with size classes drawn from a Gaussian quadrature the σ_i are irrational, so the $p(p + 1)/2$ conditions cannot be satisfied simultaneously by any choice of resolution. Type 4 is available as an option and is correct for one component, where the single core coincides with the reduced unit by construction.

The defect had been invisible to every existing test because the resulting error vanishes in precisely the monodisperse limit where those tests looked.

Under-resolved narrow wells. Comparison with an analytic adhesive-sphere solution showed errors that are governed not by the number of grid points per particle diameter but by the number *across the attractive well*:

points across well	max. relative error
2	0.916
4	0.324
8	0.142
16	0.068
32	0.035

This is ordinary first-order convergence in $\Delta r/\delta$ and not a defect, but the default discretisation violates it badly: at 100 points per diameter a well of width $\delta = 0.02\sigma$ spans two grid points and the structure factor is in error by ninety per cent, without any warning. We recommend $\gtrsim 30\sigma/\delta$ points per diameter for any potential with a narrow feature, and note that this requirement applies equally to square-well and piecewise-constant potentials.

4.7 Internal consistency

The following are not independent checks but constrain the implementation: reduction of the polydisperse construction to a single size class reproduces the original one-component routines bit-identically; the extended Rogers–Young closure reduces to Rogers–Young to 2.5×10^{-16} when its additional parameter vanishes; the hard-core double-Yukawa potential agrees with the three-Yukawa form to $\sim 10^{-16}$ under the stated reparameterisation; and the exact intensity reproduces brute-force integration of $\langle |F|^2 \rangle$ in the dilute limit (ripple 0.041 against 0.038).

5 Results

5.1 How many size classes are needed

For a charged polydisperse Yukawa system at $s = 0.2$, going from one class to three changes S^M by 0.54, while three to five changes it by 0.0015 — reproducing D’Aguanno & Klein’s claim [4] independently and confirming that polydispersity is a large effect which three moment-matched classes already capture (Figure 1).

5.2 Comparison of the approximation schemes

We computed all six schemes alongside the exact multicomponent result over a grid of relative polydispersity $s = 0.05$ – 0.40 and volume fraction $\varphi = 0.05$ – 0.40 for four systems: hard spheres and adhesive hard spheres ($\tau = 0.3$, $\delta = 0.02\sigma$) in the Percus–Yevick closure, and a square-shoulder fluid ($\varepsilon = +1k_BT$, $\delta = 0.1\sigma$) and a hard-core Yukawa fluid (screening length 0.5σ , contact strength $2k_BT$) in the hypernetted-chain closure. That is 120 state points, every one converged and screened for physicality. The error measure is $\max |I_{\text{approx}}/I_{\text{exact}} - 1|$ over $0.3 \leq Q\sigma \leq 20$, with five structure-factor classes and sixty form-factor classes. The narrow-well systems were computed at the resolution their well width demands (§4.6): 300 points per diameter for the square shoulder, 1500 for the adhesive spheres.

5.2.1 Three systems share a pattern; the Yukawa fluid does not

Note on the sign convention. The implementation writes the pair potential as $u(r) = \varepsilon$ for $\sigma < r < \sigma + \delta$ and forms $\exp(-u)$, so a POSITIVE ε gives $\exp(-\varepsilon) < 1$ and is REPULSIVE — a square shoulder. Attraction requires $\varepsilon < 0$. The survey was run at $\varepsilon = +1k_BT$ and is therefore a shoulder throughout; the calculations are unaffected, but every description of this row as a “square well” was wrong and has been corrected. Before submission, consider adding a genuinely attractive row ($\varepsilon < 0$): an attractive well and a repulsive shoulder are different physics, and the adhesive-sphere row is the only attractive system currently in the survey.

For the hard-sphere, square-shoulder and adhesive-sphere fluids the same three schemes divide the (s, φ) plane between them, in a consistent arrangement (Table 6): the scaling approximation at low polydispersity and low volume fraction, the local monodisperse approximation through the middle of the plane and increasingly as s grows, and the van der Waals one-fluid scheme along the high- φ edge. The boundaries move between potentials — the scaling region shrinks as attraction is added, and vanishes altogether for the adhesive spheres — but the arrangement does not. In all three, the decoupling and partial-structure-factor schemes are never the best choice at any state point.

The hard-core Yukawa fluid behaves differently in both respects. There the decoupling approximation is the *best* scheme at 21 of 30 points, and every scheme carries a median error between 29 and 58 %. That is a statement about which approximation is least bad rather than about any of them being usable: a parameter extracted through any of these schemes for such a system carries a systematic error of tens of per cent.

This is not an artefact of the particular Yukawa parameters. Varying the screening length over a fivefold range and the contact strength by a factor of two and a half leaves both halves of the statement intact: decoupling is the best scheme at every state point tested, and the median error across all six schemes stays between 0.55 and 0.79 throughout (Table 5).

Table 5: The Yukawa behaviour is insensitive to the interaction parameters. λ is the screening length in units of σ and K the contact strength in $k_B T$; the median is taken across all six approximation schemes.

λ	K	s	φ	best scheme	median
0.5	2.0	0.20	0.20	decoupling (0.42)	0.60
0.5	2.0	0.30	0.30	decoupling (0.35)	0.79
0.2	2.0	0.20	0.20	decoupling (0.18)	0.55
0.2	2.0	0.30	0.30	decoupling (0.24)	0.76
1.0	2.0	0.20	0.20	decoupling (0.39)	0.70
0.5	5.0	0.20	0.20	decoupling (0.44)	0.73

Two further combinations — the longest-ranged and the strongest attraction, both at $s = \varphi = 0.30$ — failed to converge. That is expected on approaching a spinodal rather than a defect, but it does mean the Yukawa behaviour is characterised on its convergent side only.

5.2.2 Typical and worst-case accuracy

Over all 120 points the local monodisperse approximation and the van der Waals one-fluid scheme are the best choice about equally often (38 and 37 points respectively), the scaling approximation at 22 and the decoupling approximation at 21 — the latter almost entirely from the Yukawa system. The partial-structure-factor scheme wins twice. Within-5 % rates over all points are 35.8 % for vdW1, 31.7 % for both the local monodisperse and scaling approximations, 16.7 % for the monodisperse approximation and 11.7 % for decoupling and partial S_{ij} .

Typical and worst-case accuracy do not agree, and the choice between them depends on the application (Table 7). For hard spheres the local monodisperse approximation has the smallest median error but the van der Waals one-fluid scheme the smallest worst case, by a factor of nearly six. Reporting only one of these would mislead.

5.2.3 The decoupling approximation

The decoupling and partial-structure-factor schemes track each other closely throughout, which is expected: both replace the distinct pair correlations by a single common structure factor. Measured against the best scheme available at the same state point, the decoupling approximation is typically worse by a factor of

Table 6: Scheme with the smallest maximum relative error at each state point. Three systems share an arrangement; the Yukawa fluid is the exception. “dec” is the decoupling and “part” the partial- S_{ij} scheme.

s	hard sphere, φ					square shoulder, φ				
	0.05	0.10	0.20	0.30	0.40	0.05	0.10	0.20	0.30	0.40
0.05	scal	scal	scal	scal	scal	scal	vdW1	vdW1	vdW1	vdW1
0.10	scal	scal	scal	vdW1	vdW1	scal	vdW1	vdW1	vdW1	vdW1
0.15	scal	scal	LMA	LMA	vdW1	scal	LMA	LMA	LMA	vdW1
0.20	scal	LMA	LMA	LMA	vdW1	LMA	LMA	LMA	LMA	vdW1
0.30	scal	LMA	LMA	LMA	vdW1	LMA	LMA	LMA	LMA	vdW1
0.40	LMA	LMA	LMA	LMA	vdW1	LMA	LMA	LMA	LMA	LMA

s	adhesive HS, φ					Yukawa, φ				
	0.05	0.10	0.20	0.30	0.40	0.05	0.10	0.20	0.30	0.40
0.05	vdW1	vdW1	vdW1	vdW1	vdW1	scal	scal	dec	dec	dec
0.10	vdW1	vdW1	vdW1	vdW1	vdW1	scal	scal	dec	dec	dec
0.15	vdW1	vdW1	vdW1	vdW1	vdW1	scal	scal	dec	dec	dec
0.20	vdW1	vdW1	LMA	LMA	vdW1	dec	scal	dec	dec	dec
0.30	LMA	LMA	LMA	LMA	vdW1	dec	dec	dec	dec	dec
0.40	LMA	LMA	LMA	LMA	vdW1	dec	dec	dec	part	part

Table 7: Median and worst-case error over the thirty state points of each system. Best in each column in bold.

Scheme	hard sphere		square well		adhesive HS		Yukawa	
	med.	worst	med.	worst	med.	worst	med.	worst
monodisperse	0.181	16.79	0.212	1.48	0.099	0.76	0.568	0.96
decoupling	0.558	286.07	0.716	28.22	0.167	5.92	0.287	0.42
local monodisperse	0.059	17.06	0.071	1.50	0.068	0.76	0.566	0.95
partial S_{ij}	0.558	286.11	0.717	28.22	0.167	5.92	0.290	0.42
scaling	0.072	66.42	0.129	6.36	0.059	1.71	0.417	0.79
vdW1	0.085	11.87	0.088	2.00	0.044	0.61	0.575	1.11

System	median ratio	worst
hard sphere	13.0	48.4
square well	14.8	43.2
adhesive HS	4.9	10.1
Yukawa	1.0	1.6

so its penalty falls monotonically as the interaction becomes softer and longer ranged, until for the Yukawa fluid it disappears.

This is consistent with the known reason for its failure, and with an independent result. Gazzillo *et al.* (1999) reached the same conclusion for polydisperse ionic fluids — that the extended decoupling approximation behaves poorly across essentially the whole Q range while the scaling approximation is accurate over the experimentally accessible range — and identified the cause: the decoupling approximation neglects the excluded-volume conditions $g_{\alpha\beta}(r) = 0$ inside the hard cores, because it replaces the distinct pair correlation functions by a small number of effective ones. Where the structure factor is dominated by excluded volume the approximation that discards excluded-volume detail loses most; where it is dominated by a long-ranged soft tail, it loses least. Our implementation of the scaling approximation retains both ingredients Gazzillo identifies as necessary, the volume prefactor and evaluation of the reference structure factor at a scaled argument, while our decoupling approximation uses a single common structure factor, the $\lambda = 1$ limit in their notation.

Draft note. This section has now been written five times, on 4, 9, 60, 98 and 120 state points, and each version supported a different conclusion: that the scheme ordering reverses between potentials (an artefact of the defect described in §4); that the local monodisperse approximation is uniformly best; that three schemes divide the plane; that the systems fall into three regimes including one where the choice barely matters; and the account above. The fourth of those was an artefact of having run only the eight lowest-polydispersity adhesive points, which are indeed insensitive to the choice — the completed row is not. The $s = \varphi = 0.40$ corner has now been checked and IS physical — the diagonal partial structure factors stay near 0.97, the intensity is positive throughout, the packing fraction reproduces exactly, and $\varphi = 0.40$ with a twofold spread in diameter is nowhere near jamming, polydispersity raising the close-packing threshold rather than lowering it. The factor of 286 therefore stands as a genuine failure of the approximation at an ordinary state point, not a numerical artefact.

5.3 Two cautions for the practitioner

Both of the following were established while validating this work and are easy to fall foul of.

Grid resolution is set by the narrowest feature. For potentials with a narrow attractive well the discretisation must resolve the *well*, not the particle: $\gtrsim 30 \sigma/\delta$ points per diameter. At the default resolution an adhesive-sphere calculation with $\delta = 0.02\sigma$ is in error by ninety per cent, and the resulting structure factor is smooth, plausible and wrong. In an earlier version of this comparison this made all six schemes appear uniformly poor (0.72–0.80) for that system, because each was being compared against a reference that was itself badly in error.

Convergence flags and residuals are not sufficient. The closure equations admit multiple fixed points. At one strongly coupled state two different accelerated solvers converged to residuals of $\sim 10^{-12}$ and returned $S(Q) = 17.0$ and 9.7 respectively, both with $\min S(Q) < 0$. Since a structure factor is a variance it cannot be negative, and solutions are therefore screened for physicality in addition to the residual test. Separately, the SUNDIALS KINSOL solver was observed to report success after diverging, because exponent clipping in the closure pins values at $\sim 10^{217}$ and defeats its internal stopping criterion; every solve consequently recomputes its own residual before the result is accepted.

5.4 Spurious oscillations from an inadequate form-factor grid

For a Schulz distribution with $s = 0.3$, brute-force integration of $\langle |F|^2 \rangle$ gives a residual high- Q ripple of 0.038, i.e. essentially smooth. Computing the same quantity on five moment-matched classes gives 1.91 — visible oscillations that a user could easily interpret as physical structure. Using a dense grid for the form-factor average, while keeping five classes for the structure factor, restores 0.041 (Figure 3).

The failure is not merely one of resolution: interpolating the full S^{AL} matrix onto the denser grid, rather than its smooth part alone, reproduces the oscillations at any grid density — a ripple of 2.79 even at $\varphi = 10^{-6}$, where S must be the identity.

6 Practical recommendations

- Three moment-matched size classes suffice for the structure factor, but the form-factor average needs of order forty; use $p_F \gtrsim Q_{\max} \langle \sigma \rangle s$.
- Resolve the narrowest feature of the potential, not the particle. For a well of width δ , use $\gtrsim 30 \sigma / \delta$ points per diameter; the default of 100 makes a $\delta = 0.02 \sigma$ well two grid points wide and $S(Q)$ ninety per cent wrong, without warning.
- A small residual does not mean a correct solve. The closure equations admit multiple fixed points; screen for $\min S(Q) \geq 0$.
- Solver convergence flags cannot be trusted; recompute the residual.
- Include instrumental resolution when fitting measured data. Omitting it biases the fitted polydispersity upward while leaving χ^2 acceptable.
- Report the parameter correlation matrix, not only error bars. In a representative hard-sphere fit the radius and volume fraction correlate at 0.75, so quoting them as independent overstates what the data determines.
- Near the critical point no closure of this family converges. The self-consistent Ornstein–Zernike approximation (Høye & Stell) is the appropriate theory there, at the cost of solving a partial differential equation over the whole density–temperature plane.

7 Conclusions

The exact treatment of polydispersity in structure-factor modelling is no longer expensive enough to justify approximating it by default. Combining moment-matched discretisation with accelerated multicomponent OZ solvers gives exact partial structure factors in a fraction of a second, for arbitrary pair potentials and twenty-four closure relations, and allows the commonly used decoupling approximations to be assessed against the exact result for the specific system under study rather than assumed to hold.

Open items before submission

1. §5.2: the systematic survey is DONE — 120 converged state points across four systems, replacing the four- and nine-point trials whose conclusions it overturned. Note that section has been rewritten five times, on 4, 9, 60, 98 and 120 points, and each version supported a different conclusion; do not tighten its language without adding state points first. The adhesive row is complete and a second Yukawa parameter set has been added, so the regime picture no longer rests on a single slice.

2. A fit to a real measured dataset is required; all results here use synthetic data generated from the model being fitted. This became more urgent rather than less: several defects found since — the rank-revealing linear solve returning zeros at a condition number of 10^{20} , a fit-flag reset that silently left only one parameter free, and a Gaussian resolution kernel used where dQ/Q reached 0.55 — each affected every fit made before they were corrected. Any measured-data result must be obtained after them, not before.

3. **The transform type needs a decision, not just a description.** Measured against the exact Percus–Yevick compressibility, the first-order DST-I gives $S(0)$ low by 5.27 % at $\varphi = 0.40$ against 0.036 % for the second-order DST-IV on the same grid — a factor of 148, because the hard core falls ON a type-1 node and BETWEEN type-4 nodes. Since $S(0)$ is the compressibility, this is a systematic bias in exactly the quantity φ controls, and a fit absorbs it by moving φ .

Type 4 is nonetheless NOT the default, and cannot be for the polydisperse route at all: it requires every pair core between grid points and the σ_{ij} of a moment-matched quadrature are mutually incommensurate. Decide whether to default the one-component path to type 4, and state the polydisperse limitation where a reader will meet it rather than only in the validation section.

4. **The RMSA row is now qualified and the claim should be read again.** “Exact (9/9)” held at nine state points where the Hayter–Penfold rescaling never engaged. Where it does, the rescaled volume fraction still agrees with the reference to four decimals but the curves agree only to between 10^{-14} and 3×10^{-5} , degrading with coupling strength, and one of nine fails to converge outright. Decide how much of that belongs in the table and how much in the text.

5. **The report and this manuscript keep separate bibliographies for the same papers** — daguanno1991 against DAguannoKlein1991, 36 entries in all. A correction to one does not reach the other, which is how the Carbajal–Tinoco closure carried a wrong attribution in two documents at once. All 36 already have counterparts in `ozgui.bib`; unifying them is a rename plus a switch from `thebibliography`, to be done in one pass and matched on TITLES rather than author surnames, since a surname match is what conflated two papers by the same author in the first place.

6. The introduction’s literature framing: PARTLY DONE. It now places the work against the analytic solutions, the general-purpose solvers and the scattering packages, and states the gap the software fills. That framing rests on the packages actually used during development and validation, which is not a systematic survey; check in particular whether any package now combines exact multicomponent solution with arbitrary closures, and expand the prior polydisperse OZ work beyond D’Aguanno and Klein [3].

7. Timings: DONE. Measured with `tools/benchmark.py` on a stated machine and tabulated in §3.2. Re-run that script if the hardware or the solver set changes; it writes JSON incrementally, so a failure in one block does not discard the others.

8. Author list, acknowledgements, funding, data availability.

9. AI-assistance disclosure: a statement has been DRAFTED (see §“Use of AI assistance” at the end of this document). It still needs checking against IUCr’s own policy, which may prescribe wording or placement, and the author should confirm that the description of the division of labour matches their own view of it. included.

10. Transfer to the IUCr document class.

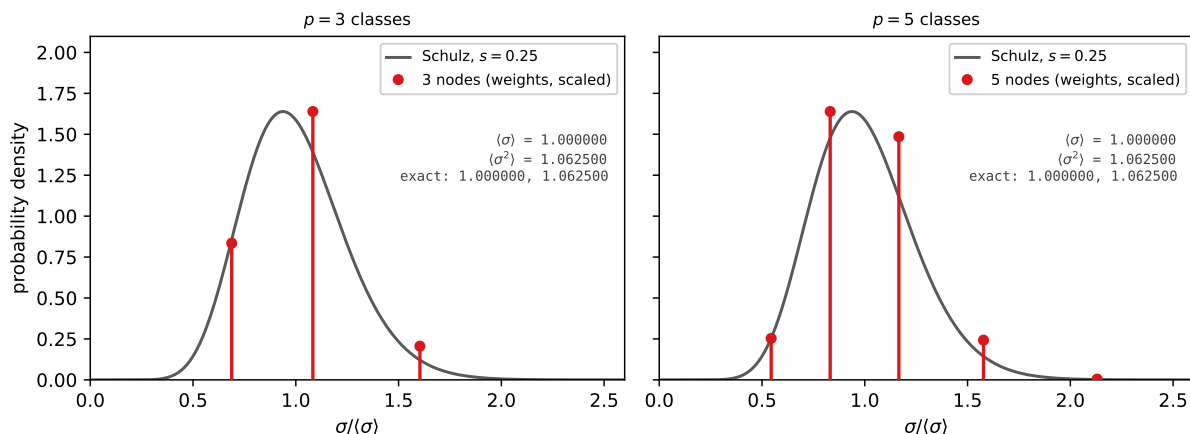


Figure 1: Moment-matched discretisation of a Schulz distribution of relative width $s = 0.25$ (solid line) into three (left) and five (right) size classes, shown as the nodes and weights of the generalised Gauss–Laguerre rule. A p -point rule reproduces the first $2p - 1$ moments exactly.

Software availability

The software is distributed with SASfit. [COMPLETE BEFORE SUBMISSION: repository URL, version tag or DOI, licence.]

Use of AI assistance

Parts of the implementation, its test suite and this manuscript were developed with the assistance of a large language model, used interactively as a programming and drafting aid throughout. Its contribution was concentrated in writing and refactoring code, constructing validation scripts against external reference implementations, and drafting and revising text. All physical content, the choice of models and closures, the design of the validation strategy, and every numerical result reported here were checked by the author; several of the errors described in §4 were found precisely because results produced with such assistance were not taken on trust.

We note this not only because disclosure is expected but because it is methodologically relevant. Software written this way accumulates plausible but unverified assertions particularly easily — a convergence flag read as evidence of a solution, an approximation assumed cheap because it looked cheap, an identical-cores defect that produced entirely reasonable-looking structure factors. The emphasis throughout this paper on validating against independent implementations, and on checking residuals rather than trusting solver reports, is a direct response to that.

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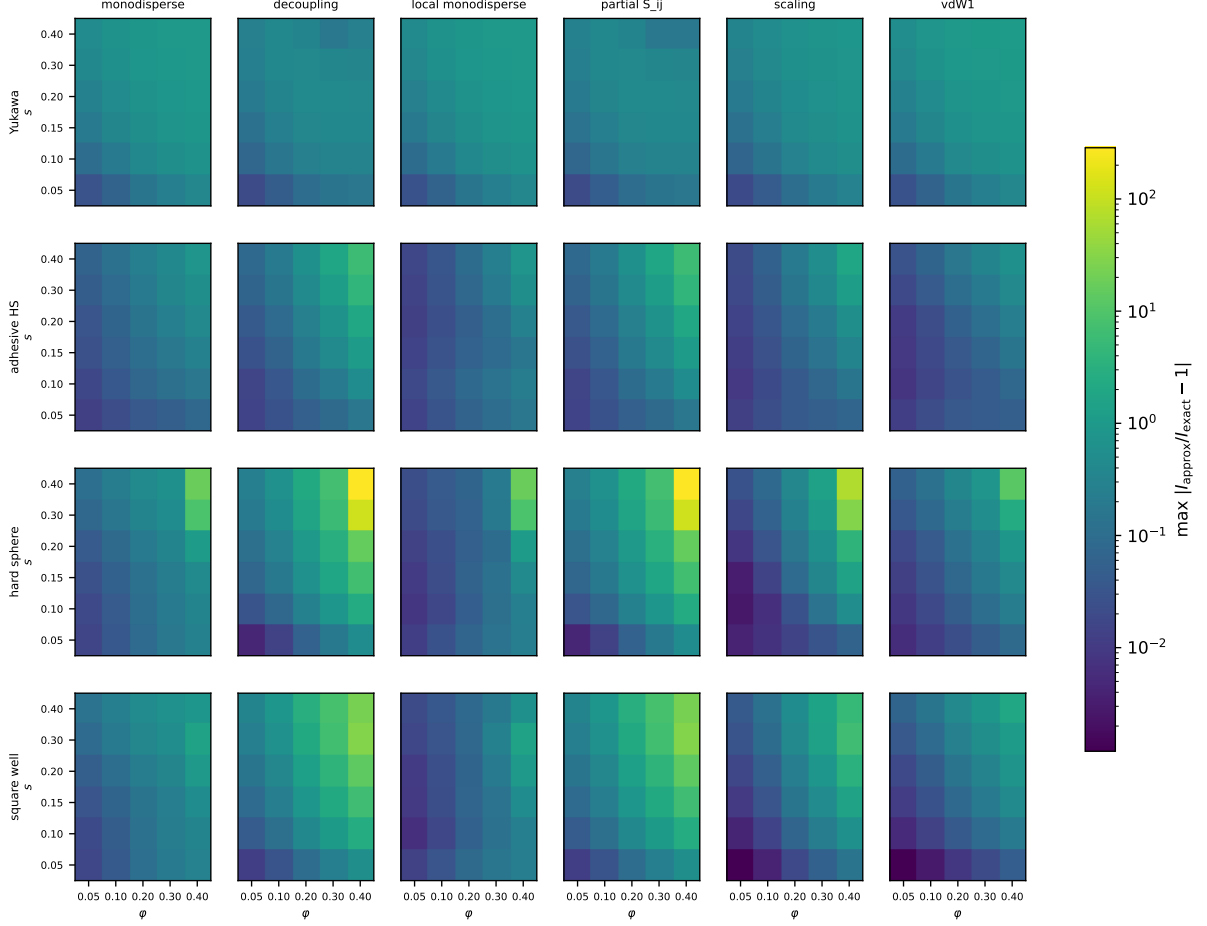


Figure 2: Maximum relative error of each approximation scheme against the exact multicomponent result, over the (s, φ) plane, for hard spheres (top) and a square-well fluid (bottom). Sixty state points, all converged. The colour scale is common to all panels and logarithmic, so the panels are directly comparable. The decoupling and partial-structure-factor schemes (second and fourth columns) fail together and most severely at high polydispersity and high volume fraction, reaching a factor of 286 for hard spheres at $s = \varphi = 0.40$; the local monodisperse and van der Waals one-fluid schemes remain usable across most of the plane. Generated by `tools/section53_survey.py`.

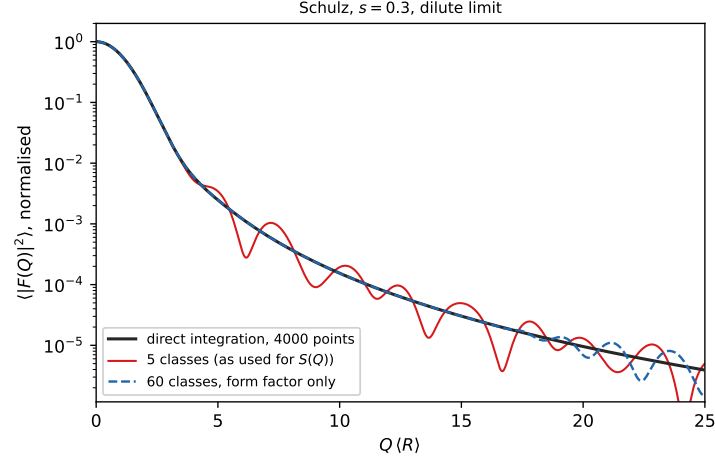


Figure 3: Spurious form-factor oscillations from an inadequate discretisation of the form-factor average, for a Schulz distribution of width $s = 0.3$ in the dilute limit. Direct integration with 4000 points (black) is essentially smooth; evaluating on the same five classes used for the structure factor (red) superposes five ringing form factors. Sixty classes for the form-factor average alone (blue, dashed) restores the correct envelope.

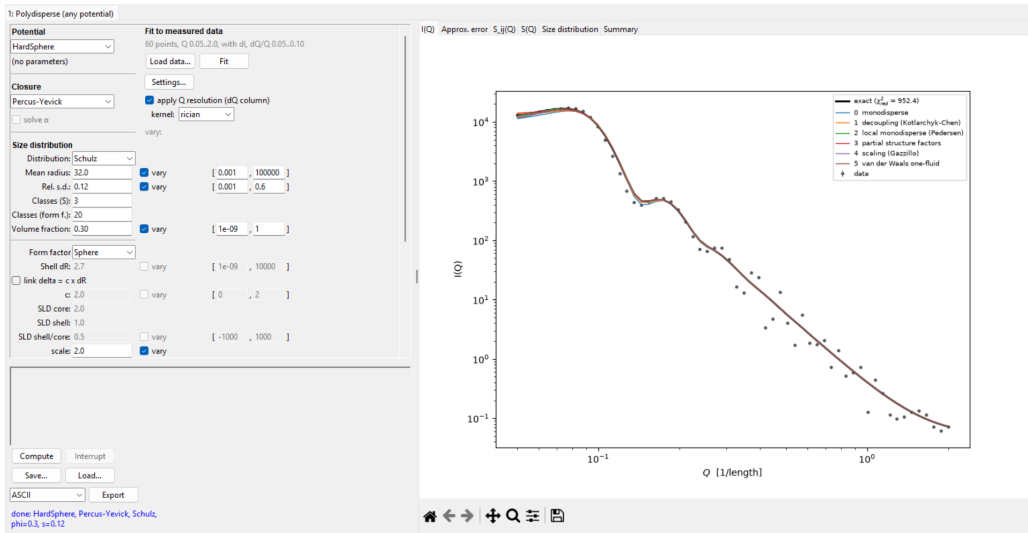


Figure 4: The polydisperse interface. Potential and closure parameter fields are generated by introspection of the corresponding setter; the size-distribution block carries separate class counts for the structure factor and the form-factor average (§2.4).

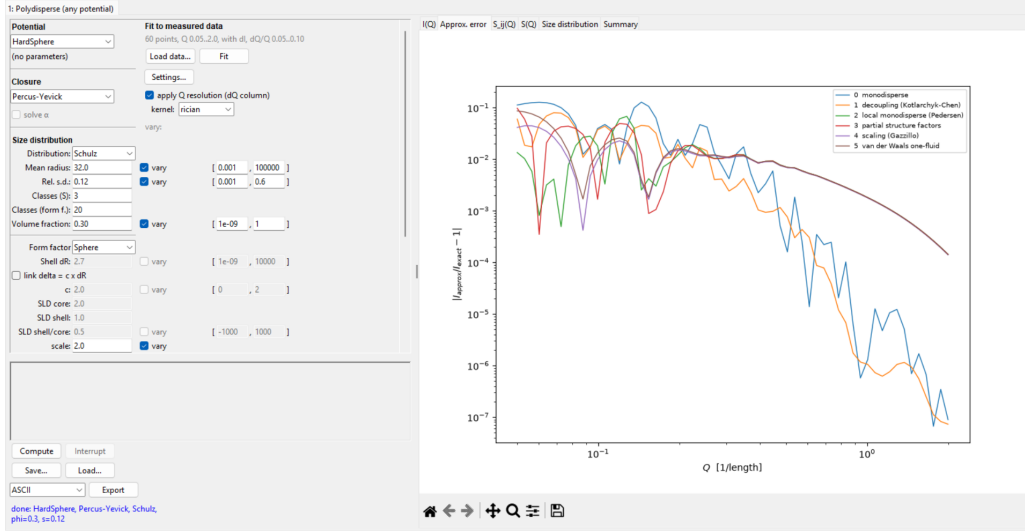


Figure 5: The approximation-error panel, showing the relative error of all six schemes against the exact result for the system currently under study. This is the comparison the software is intended to make routine.

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